

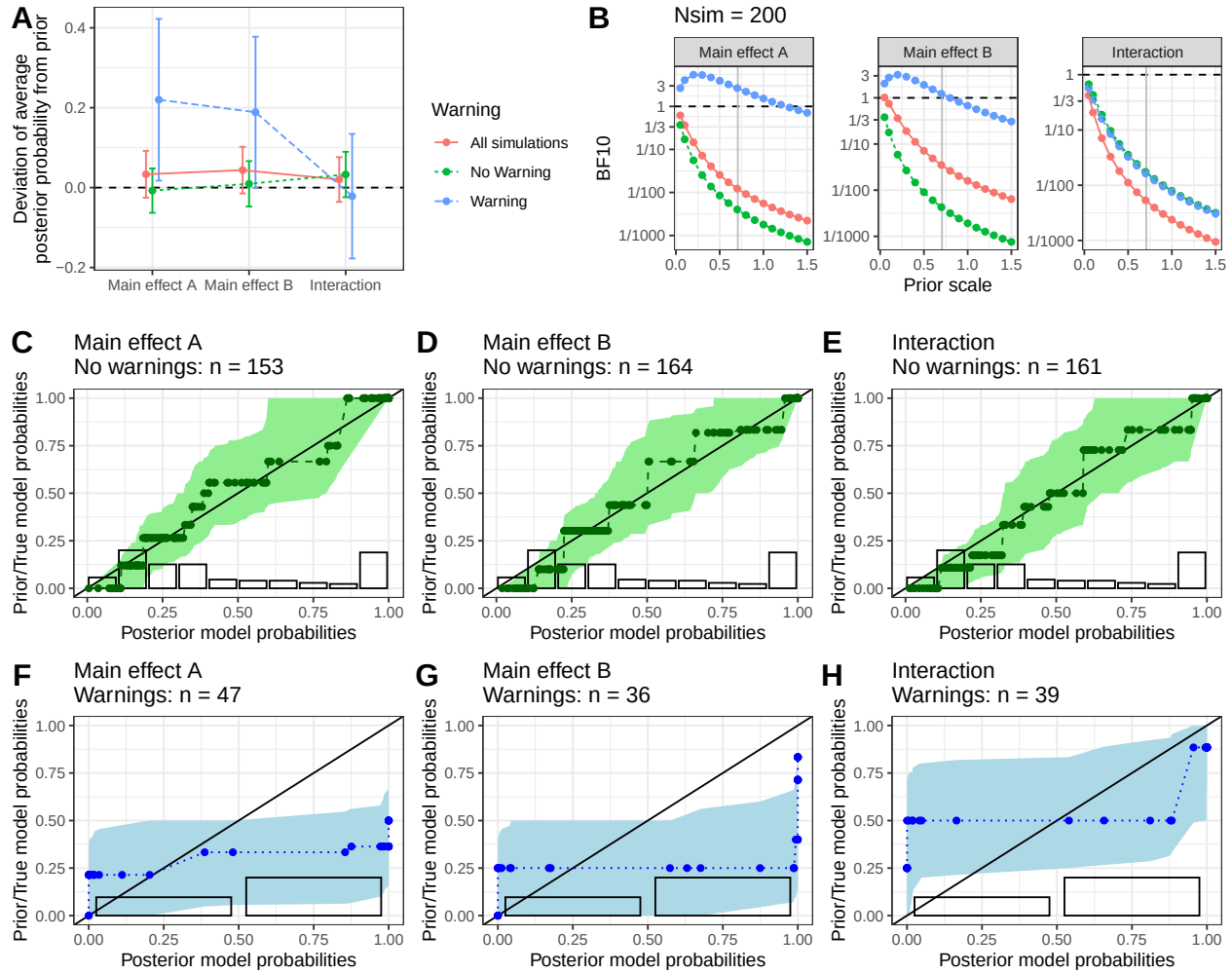


Figure 1. Results from simulation-based calibration (SBC) for a 2 x 2 repeated measures design. A) The average deviation of the posterior model probability from the prior (with 95% confidence intervals) is shown for the main effects A, B, and their interaction. Data is shown for all simulations and separately for those without / with a warning message from the bridge sampling algorithm. The horizontal broken line indicates no deviation; deviations from this line indicate bias. B) Bayesian t-tests of whether the posterior probability for H1 deviates from the prior probability, with sensitivity analyses for the prior scale. The vertical grey line indicates a default Bayes factor. C-H) Reliability diagrams: the prior model probability is plotted as a function of the estimated posterior model probabilities. Error bands show 95% consistency bands. Results show that when no warning message was issued (C-E) prior model probabilities do not deviate from posterior model probabilities, providing evidence for accurate estimation, but that when a warning message was issued (F-H) prior and posterior model probabilities do not match, providing evidence for biases.

showing default Bayes factors of $BF_{01} > 10$ for the null for the two main effects and the interaction. A prior sensitivity analysis showed that these results were stable across different prior scales, being present already for very small prior scales, and becoming stronger for larger prior scales. These results suggest no evidence for overall bias in the Bayes factor estimates when no warning message is shown. For simulations where there was a warning message from the bridge sampler, the results were less clear, with sensitivity analyses showing either evidence for no bias (interaction), or unclear evidence (main effects).

We moreover used reliability diagrams (Dimitriadis et al., 2021) to investigate whether the distribution of estimated posterior model probabilities is calibrated. If the estimates are well calibrated and accurate, posterior model probabilities should accurately predict the true model, which would be visible in Figure 1C-H in estimates aligned to the diagonal. Indeed, it is visible in Figure 1C-E that when no warning message was issued by the bridge sampler prior and posterior model probabilities did match, suggesting accurate estimation. However, in cases where the bridge sampling algorithm issued a warning message, the estimates deviated from the diagonal, suggesting bias in the resulting Bayes factor estimate (see Fig. 1F-H).

Importantly, additional analyses showed that we observed warnings from the `bridgesampling` package only for data sets that required small step sizes and large treedepths to explore with Stan’s HMC sampler and which in turn required a lot of computation time to fit. This implies that the posterior geometry was highly irregular, which would explain problems faced by bridge sampling. In particular, the problematic data sets tended to result in small fitted residual standard deviations and/or some random effect standard deviations close to zero. When faced with such posteriors, a user may consider reparametrizing the model to avoid the computational difficulties and we would expect a successful reparametrization to also resolve the `bridgesampling` warnings. E.g. for the cases with low residual standard deviations, the data are actually highly informative of the random effects

which results in weak non-identifiability between the overall intercept/effect and random intercepts/effects (see Ogle & Barber, 2020, for more discussion). The models worked well when dropping the random effect for the interaction. Also, in some (but not all) of the cases, enforcing a sum-to-zero constraint on the random intercept and each random effect results in improved sampling and no warnings from bridgesampling.

Design 2: One 2-level factor with crossed random effects for subjects and items

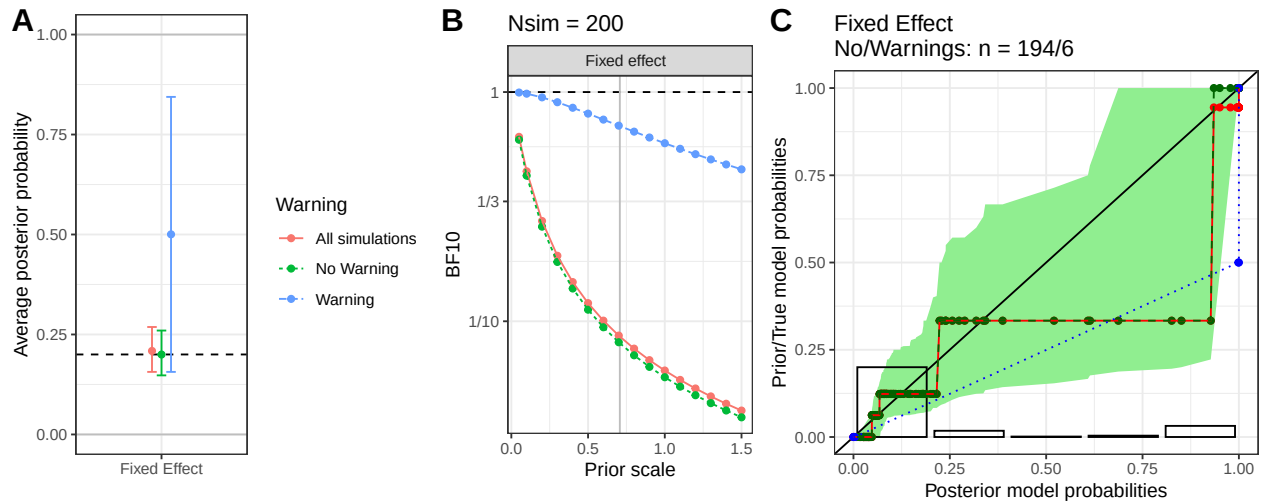


Figure 2. Latin square design with one 2-level fixed factor and random effects for subjects and items. A) The average posterior model probability (with 95% confidence intervals) is shown for the fixed effect. Data is shown for all simulations and separately for those without / with a warning message from the bridge sampling algorithm. The horizontal broken line is the prior probability for the H1; deviations from this line indicate bias. B) Bayesian t-tests of whether the posterior probability for H1 deviates from the prior probability, with sensitivity analyses for the prior scale. The vertical grey line indicates a default Bayes factor, providing evidence for no bias. C) Reliability diagram: the prior model probability is plotted as a function of the estimated posterior model probabilities. Error bands show 95% consistency bands (only for simulations without warning). Results show that prior model probabilities do not deviate from posterior model probabilities, providing evidence for accurate estimation.

Next, we studied a Latin-square design with a 2-level fixed factor and crossed random